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Data Biasing Removal with Blockchain and Crowd Annotation

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Abstract

In today's world of internet marketing product review plays a crucial role. Often E-Commerce companies are accused of manipulating with the data and it suffers from biasing. Biasing leads to wrong customer's view creation thereby prediction for a top rating product fails. This study shows how crowd annotation prevents this biasness and thereby design a smart contract which ultimately will help to eliminate the data manipulation. The smart contract is deployed and tested to see if data can be saved in the run time storage memory regions of the Ethereum blockchain, and the test results are positive. In future this will help to get the unbiased data and better machine learning results, which will ultimately be beneficial for the economy.

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Keywords: Machine Learning; Biasing; Crowd Annotation; Blockchain; Smart Contract

1. Introduction

Data plays a key role in both predictive and prescriptive analytics. In data analytics it is assumed that the data is free from biasing. Biasing does exist in every domain as in case of clinical data, where starting from population selection to research models' development biasing gets induced as discussed in [5]. There are three basic categories of biasing mainly (a) Historical and social that directly comes from the data (b) Emerging and popularity which comes from the algorithm and (c) behavioural and presentation bias, which is caused mainly through user interaction

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as discussed in [3]. This paper aims to address category (a) and (c) i.e., social data as customers are involved and customers are an integral part of society. Behavioural and presentation bias is involved as there is virtual connection between e-commerce sites and its users. As a case study the paper takes e-commerce sites into account. AI driven e-business is the talk of today as mentioned in [13]. But all major literatures reveal the laguna of AI if the training set is biased and this results in sufficient increase in the false positive numbers. The consequences are heavily burdened for the economy as well as society. Sometimes it is seen that the review topic or review content of e-commerce sites are not matching with the rating which customers have provided. It is widely possible that customers or users do not understand the importance of rating as a result a low-quality product may get a high rating and future customers get attracted to the product unknowingly as AI Engine also predicts the wrong product. Economic loss is for those hard-working manufacturers who do not compromise on quality but in turn gets a poor rating even if their review content is good. Blockchain based feedback system is one of the solutions well addressed in [12,14]. This encourages the requirement for the availability of a web scrapping data set where customer review title, review comment and rating are present in the data set. This data set [17] is encouraging and the paper tries to provide answers to the following questions

(a) Does there a correlation between the Review Content and sentiment analysis annotation?
(b) Does Review Content match with Rating?
(c) From the Sentiment Analysis can the rating of a product be predicted?
(d) How can be it ensured that Review Content has not been manipulated and thereby design a mechanism which will not allow any changes in the review system?

In this paper a data set is taken and crowd annotation is performed to check whether sentiment analysis is matching with the review count or rating value. Kappa Score (Appendix-A) value calculated in the project indicates that there is random agreement among crowd workers. The whole data set is put under regression and linear regression best fits in. Sentiment analysis methodology does not correlate the rating values but crowd annotation does. However, crowd biasing does exist and addressing of crowd biasing is beyond the scope of this paper. The rest of the paper is organized as follows. Section 2 presents the related work considering prediction and biasness. Section 3 describes the proposed methodology to answer the research questions. In section 4, smart contracts development and performance are presented. Finally, section 5 concludes our work and proposes the future work.

2. Related Works

AI Contentfy team rightly points out that AI generated content is going to change the market of tomorrow. Accordingly, they are of the view that large amounts of data such as customer behavior and product information are fed into AI Model and inference is generated. If the data is incomplete, inaccurate, or biased, the resulting content may be flawed and less effective which is the problem that are faced by major e-commerce sites [16]. If only past ratings are being considered, it never gives accurate result. So, researchers have opted for user-feature matrix and found that matrix factorization method works much better while taking the user-feature matrix into account [1]. Some Literature rightly points out the type of biasing involved while customers give review. Social influence and selection bias can be removed by random sampling and found that enthusiastic reviewers aka self-motivated reviewers tend to give lower star ratings and authors are of the conclusion that reviews received from self-motivated are good quality values received [2]. Some past works have seen implementation of different algorithms on biased data and found that maximum algorithms give poor result compared to that of annotated data. However, biasness may exist with annotated data too [3]. Researchers have identified certain methodologies like Fairness Metrics valid for (individual or group), NLP and understanding sentiments of texts or sentences, mitigation of bias and unfairness in tabular data. Researchers are of the view that more research should be carried out in crowdsourced biasness removal, database schema design, algorithmic mitigation to avoid these problems. Data science researchers have seen that bias can occur in the planning, data collection, analysis, and publication of research. So thus, it is seen that in the training phase biasing can be inducted [5]. Researchers have shown through synthetic data set generation that those actual ratings shown by Amazon is not the actual representation. The study shows that there are evidences of rating differences by gender, age, ethnicity, income, and mobile device used for shopping. So true rating is never depicted in e-commerce sites [6]. An interesting fact is shown that what is the impact of online product review on consumer purchase through eye-tracking. Interestingly it has been found that negative comments get more attention compared to the positive one. Consumers cannot identify false information or review. So, e-commerce sites need to consider the negative comments such that negative comments get reduced specially if they are targeting female

customers [7]. Some work has taken place in cross cultural factor analysis of online review system in e-commerce site. Researchers have identified significant predictors and cultural perspective as another factor and then they performed a quantile regression across the entire distribution of the count of helpful votes. It is believed that distortion in this way will be reduced up to a certain extent [8]. There has been testing with review system in many surveys conducted. In a work it is seen where researchers have surveyed more than hundred respondents and have concluded that brand commitment mediates the association between brand love and brand trust and between brand trust and brand loyalty. There is cognitive dissonance among buyers which should be taken care of from the retail side such that the brand loyalty persists [9]. Researchers have taken a case study and have analyzed the effect of reviews on online purchase in specific regions and concludes that reviews have a positive effect on buying decisions and they are of the opinion that online reviews affect consumer attitudes and can significantly influence e-commerce sale in India [10]. There has been an examination of secondary data from Amazon and researchers have analyzed to get a J-shaped distribution with more positive feedback than negative feedback. In the review system positive and negative are of equal distribution. In the feedback system of two types of biases are noted (a) Purchasing bias and (b) under reporting bias where it is seen of that customers who have made their views to purchase a product can give review and similarly those customers who give only positive and negative reviews have tendency to write reviews and on the other hand those have neutral bend of mind hardly gives review [11]. A brief use case of feedback system development with blockchain is introduced conceptually by some papers. The major use cases of this Blockchain system includes feedback system where the secrecy of the participants is maintained totally. According to the paper encryption process is attached with Blockchain system [12]. The survey indicates that major areas of Blockchain research is spread across several areas like supply chain, business, healthcare, IOT, privacy and data management. The advantages and disadvantages of Blockchain is well addressed in [13]. A Blockchain voting system is introduced in [14] where there is the design of an excellent voting feedback system whose parameters get adjusted. Authors are of the opinion that malicious votes can be suppressed by the feedback mechanism.

3. Methodology

Here, the proposed methodology is represented to develop a method which would prevent the manipulation of review data. First, the annotation part is seen and then in the second part smart contract based Blockchain Technology is used for entering the review of data. As already mentioned in Introduction part the paper shall deal with three research questions: -

- Does there a correlation between the Review Content and sentiment analysis annotation?
- Does Review Content match with Rating?
- From the Sentiment Analysis can the Rating of a Product be predicted?
- How can be it ensured that Review Content has not been manipulated and thereby design a mechanism which will not allow any changes in the review system?

3.1 Dataset Information

The data has been collected from data world [17] and initial data set did consist of the following. The data set has been prepared by data world in collaboration with Prompt Cloud and Data Stock. As the data are a result of crawling information so unique id and Billing id has been encoded to hex code by the crawler authority.

Table 1. Dataset Information

Name	Description of Units
Unique id	Hex code representing Transaction
Crawl Timing	Time when data was crawled in yy-mm-dd and time in hour: minutes: second
Billing Id	Hexadecimal encoded value
Rating	Numerical value blank in this case where perhaps some internal calculation algorithm will give the final rating of the product

Review Title	Each Review is associated with review title basically string type
Review Rating	Rating given by Customer within range of 1-5
Review date	Month, Date, Year type Integer
User Id	Encoded Hexadecimal Code for Security purpose
Brand	String Type showing exact Brands for which the product review came
Category	String type indicating for each product it belongs to which category for example grocery, skin care etc.
Sub Category	String type indicating specific product category for example frozen, skin care, personal care etc.
Product Description	String Type specific Brand specific item for example Shea Moisture and description about the product
Asin	Varchar type or String Type consists of both alphabets and digits basically a label for a particular product
Url	Url of the website that was crawled or crawled pages that came in between the main index page.
Review Content	String mainly paragraph helps to identify the sentiment of the reviewer about the product
Verified Purchase	Flag type yes/no basically in E-commerce site review should be given by a customer who has purchased and only those reviews are counted who have purchased.
Helpful Review Count	Number of persons who were benefited from the review.
Manufacturer Response	Blank generally e-commerce sites send review to manufacturers.

3.2 Data Preprocessing

As mentioned in the data as shown in Table-1 only a few fields like Review Title, Review Content and Rating were considered. Rest of the fields were eliminated out. As the table consists of only 50 sampled data so no data were eliminated out. However, it was checked that the date of purchase was less than crawl time. Purchase verification was not taken in this experiment.

3.3. Sentiment Analysis

The sentiment analysis was performed in phases. For Sentiment Analysis, VADER Sentiment Analysis, was done where VADER means Valence Aware Dictionary and sentiment Reasoner. Once polarity value for a given sentence was calculated both for “Review Tite” and “Review Content” field separately it was passed through a Polarity function as per the following rule: -

$$S(\text{Polarity}) = \begin{cases} 0 & \text{if } 0.05 <= \text{Polarity(score)} <= 0.30 \\ 1 & \text{if Polarity(score)} > 0.30 \\ -1 & \text{if Polarity(score)} < 0.05 \end{cases}$$

3.4 Correlation Value

The correlation values between "Review Content polarity values," "Rule-based sentiment analysis for Review Content," "Rating provided by users" were calculated as shown in Table-2. An interesting result is observed that "Review Content Polarity" just matched 60% with the "Rule Based Sentiment Analysis for Review Content". Thus, the Review content Polarity and Rating matched 30% and rule-based sentiment analysis just matched 39.22% with Ratings provided by users. This indicates that Review content and Ratings did not match and clearly indicates that biasing exists. However, compared to "Review Content" "Review Title" was interpreted with a better performance by VADER. The correlation between "Review Title Polarity" and "Rule based Sentiment Analysis" was found to be 77.3% whereas Rating match with both sentiment polarity and rule-based polarity for Review Title was found to be 47% and 39% respectively. So, it is concluded that direct sentiment analysis and Rating prediction is not possible. This is depicted in Table 3.

Table 2 Correlation Value between Review Content Polarity, Rule Based Sentiment Analysis and Rating

Parameters	Review Content Polarity	Rule Based Sentiment Analysis	Rating
Review Content Polarity	1.0000	0.6090	0.3063
Rule Based Sentiment Analysis	0.6090	1.0000	0.3922
Rating	0.3063	0.3922	1.0000

Table 3 Correlation Value between Review Title Polarity, Rule Based Sentiment Analysis and Rating

Parameters	Review Title Polarity	Rule Based Sentiment Analysis	Rating
Review Title Polarity	1.0000	0.7735	0.4709
Rule Based Sentiment Analysis	0.7735	1.0000	0.3930
Rating	0.4709	0.3930	1.0000

3.5 Annotation through Crowdsourcing

Out of all the data fields the first research question was investigated. A new table was taken with Product id (Asin), Product Name, Review Rating, Review Title, Review Content, Verified Purchase. Sentiment was analysed with just Review Content. Thus, sentiment was added as a new annotated field. The sentiment was annotated based on 3 criteria. A set was taken comprising of {-1,0,1} where 1 indicates perfect positive feedback, 0 means neutral feedback and -1 means negative feedback. The first research question that is whether Review Rating had a perfect correlation with Review Title or Review Content as well as sentiment analysis. The data was sent to two volunteers who are well accustomed with online shopping review and rating. However, they were asked both first and second research questions together to annotate. The whole preprocessing, processing and annotation scheme is explained as shown in Fig.1. and Fig.2 respectively. The manual crowdsourcing data was analysed with two crowdsourced volunteers and kappa score was found to be 0 which means that there is random agreement between the raters. Detailed calculation of crowd annotation has been shown in Appendix-A. Though it is not possible to calculate the correlation between Review Content as this is a non-quantifiable variable but Review content is directly proportional to sentiment value. So, it is possible to calculate the correlation matrix between Sentiment value and Review rating as shown in Table 4.

Table 4 Correlation Matrix with Crowd Annotation

Parameters	Review Rating	Sentiment
Review Rating	1.000000	0.901757
Sentiment Analysis by Crowd	0.901757	1.000000

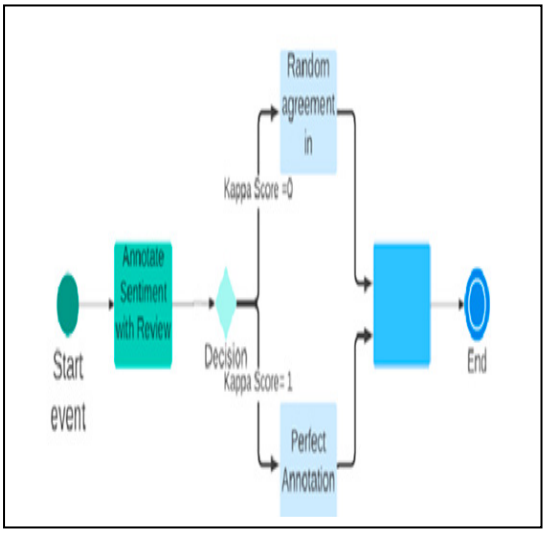


Fig.1. Data Preprocessing and Data Processing Step

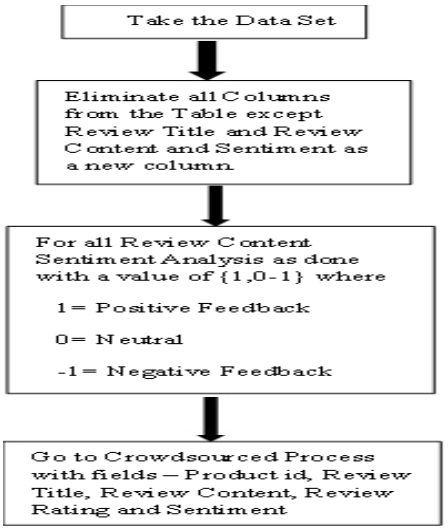


Fig.2. Annotation Process with Crowdsourcing

3.6 Prediction Methodology with Machine Learning

As seen from Section-3.2 there is a high correlation value between sentiment analysis and Review Rating while taking crowd annotation into account. As sentiment analysis was taking values from the set {1,0, -1} there was a high probability that logistic regression would yield some result with the data. But to predict ratings with sentiment analysis Linear Regression outperformed Logistic Regression. The result is shown in Table- 5.

Table 5. Regression Analysis Result

Model with Dataset	Accuracy
Linear Regression	0.8654
Logistic Regression	0.6923
Polynomial Regression	0.5489

So thus, we conclude that Crowd annotation methodology outperforms sentiment analysis methodology. This crowd annotation methodology is thus helpful in reducing bias however Linear Regression result of 86% indicates that 14% may be considered as crowd bias [15]. Result of the Linear Regression Analysis is shown in figure 3.

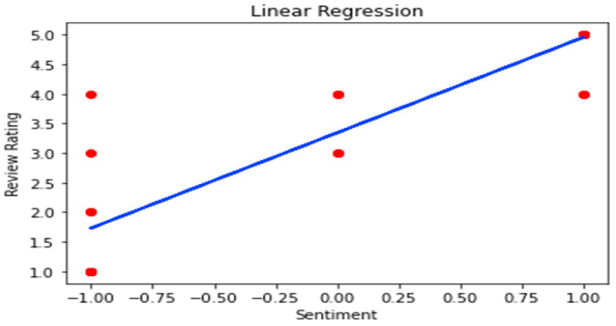


Fig.3. Result of the Linear Regression Analysis

3.7Methodology to eradicate Biasness

In this paper it has already been illustrated how crowd annotation prevents biasness. Another method we propose here is smart contract based Blockchain system. A smart contract is written in the Ethereum Platform to enter the review of the product. Blockchain system in brief is described below.

3.7.1 Blockchain

A Blockchain is a distributed database or ledger shared among shared network nodes Once entered data cannot be changed or we can say immutable data. Blockchain Process is shown in figure 4. So, no third party is required and the

block stores all the information and any change made in the content of a block is caught as the hash value of the block changes with timestamp. This is well illustrated in the Figure 6.

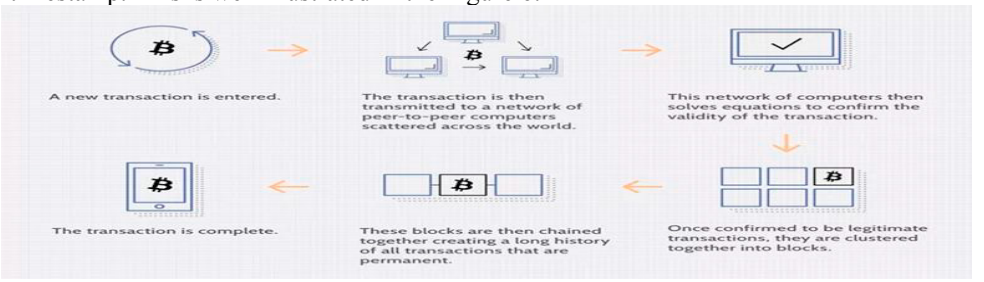


Figure.4. Blockchain Process Source: Investopedia | Pictorial representation of how blockchain technology works

3.7.2 Smart Contract

A smart contract is a self-executing computer program that automatically executes without the intervention of 3rd parties. Smart contracts are generally written in high level languages like solidity which runs on Ethereum Virtual Machines (EVM). Blockchain just act as a platform where smart contracts output shall reside.

3.7.3 Methodology Used

A two-level security system has been implemented with Smart Contract. In the first case user_id is automatically updated such that a customer can give review for a product only once. A Central authority monitors if there is malicious transaction. The whole process diagram is shown in Figure-7. As soon as any value which is stored in a node gets changed the hash value of the node changes. For hash conversion Blockchain basically uses SHA-256. The whole process works as a DApp (Decentralized Application). The output of the Review result is stored in a file which can be processed by the ML Engine to process and predict the rating for a given sentiment. Layout Diagram of the Ethereum Contract used in the Project is shown in figure 5.

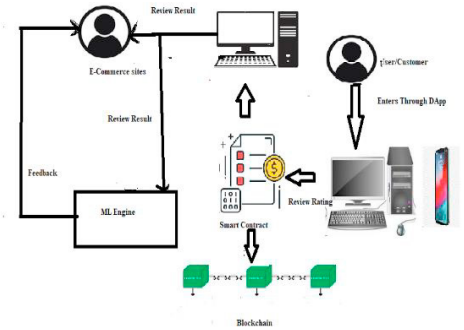


Figure.5. Layout Diagram of the Ethereum Contract used in the Project

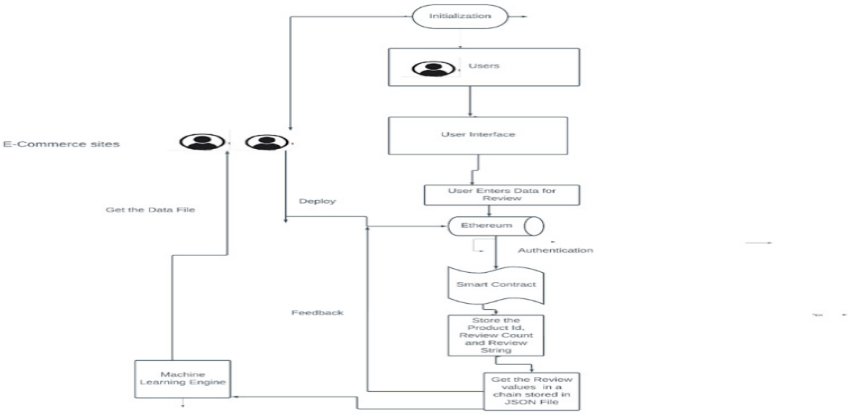


Figure.6. Process Layout of the Contract that has been developed in the Project

4. Experiment Analysis-Smart Contract Function

In this section, there is introduction of the functions required by the smart contract and deployed the smart contract. The smart contract is tested in the Solidity Remix IDE integrated development environment. The smart contract uses the Solidity language. The Solidity Remix IDE provides an online compilation environment for the Solidity smart contract. In the JavaScript VM framework, a single address is taken into consideration and ether allotment is not compulsory in this contract. The smart contract structure is shown in Figure 7. The methods are deployed and tested in Solidity Remix IDE environment. The result is shown in Section 5.

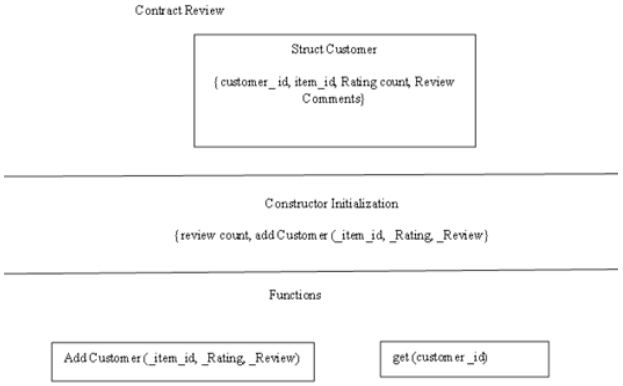


Figure.7. Smart Contract Structure Diagram

5. Result & Discussion

The whole Ethereum contract was tested on Remix where it maintained a constant accuracy of 80% with the following results as portrayed in Table 6 and Table 7 indicating run time and execution time of the smart contract. Execution Time of Smart Contract Deployed in the System is shown in figure 8.

Gas Cost: -341400

External Cost: - 227

Table 6. Smart Contract Test Result

Number of Data Entered	Number of Key Value Pairs Generated	Re-entrancy Attacks	Integer Overflow	Front Running	Gas Limit/Out of Gas Issue
10	21	No fund involved	X	X	No issue
20	41	No fund involved	X	X	No issue
30	61	No fund involved	X	X	No issue
40	81	No fund involved	X	X	No issue
50	101	No fund involved	X	X	No issue

1. Re-entrancy Attacks: - An attacker repeatedly re-enters a vulnerable contract to drain funds.
2. Integer Overflow/Underflow- Unexpected behaviours due to exceeding maximum or minimum values of integers.
3. Front Running: - Manipulating the transaction order to gain an unfair advantage.
4. Gas Limit and Out-of-Gas Issues: Poorly optimized contracts may run out of gas during execution.

Table 7. Smart Contract Run Time Result

Number of Test Cases	Passed	Execution Time
5	4	0.18s
5	4	0.16s
5	4	0.19s
5	4	0.18s
5	4	0.16s
5	4	0.16s
5	4	0.15s
5	4	0.17s
5	4	0.19s
5	4	0.18s

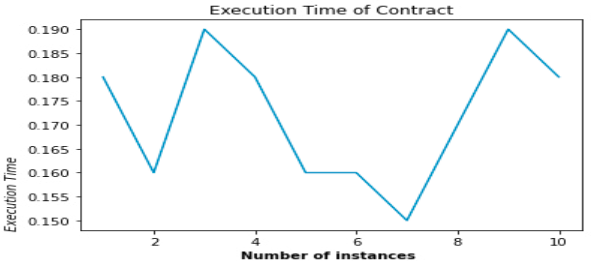


Figure.8. Execution Time of Smart Contract Deployed in the System

6. Conclusion

Thus, we see VADER sentiment analysis fails to find good correlation between review Content / title. and ratings. However, with Crowd Annotation an accuracy of 86% is achieved. And those answers all research questions. To avoid any discrepancy, it is always better to use Blockchain Contract. Blockchain contracts in this experiment attains around 80% score. It thus helps the rater to give correct data. However more testing is required to make the system a robust one. In future DApp will be used in the Review System.

Appendix A. Annotation by Volunteers

Serial No.	Product ID	Annotation by Volunteer-1 Sentiment Analysis		Annotation by Volunteer-2 Sentiment Analysis	
		Agree	Disagree	Agree	Disagree
1	B071YYMZ19	✓		✓	
2	B071YYMZ19	✓		✓	
3	B071YYMZ19	✓			✓
4	B071YYMZ19	✓			✓
5	B071YYMZ19	✓		✓	
6	B071YYMZ19	✓			✓
7	B071YYMZ19	✓			✓
8	B071YYMZ19	✓		✓	
9	B071YYMZ19	✓		✓	
10	B071YYMZ19	✓		✓	
11	B071WKSX1F	✓		✓	
12	B071WKSX1F	✓		✓	
13	B071WKSX1F	✓		✓	
14	B071WKSX1F		✓	✓	
15	B07V3978XD	✓		✓	
16	B07WDL82Z4	✓		✓	
17	B07WDL82Z4	✓		✓	
18	B07WDL82Z4	✓		✓	
19	B07WDL82Z4	✓		✓	
20	B07WDL82Z4	✓		✓	
21	B07WDL82Z4	✓		✓	
22	B07WDL82Z4	✓		✓	
23	B07WDL82Z4	✓		✓	
24	B07WDL82Z4	✓		✓	
25	B07WDL82Z4	✓		✓	
26	B007O0GQPC	✓		✓	
27	B007O0GQPC	✓		✓	
28	B007O0GQPC	✓		✓	
29	B007O0GQPC	✓		✓	
30	B007O0GQPC	✓			✓
31	B007O0GQPC	✓		✓	
32	B007O0GQPC	✓		✓	
33	B007O0GQPC	✓		✓	
34	B007O0GQPC	✓		✓	
35	B007O0GQPC	✓		✓	
36	B00FYZJPAM	✓			✓
37	B07NQN58ZC	✓		✓	
38	B07NQN58ZC	✓		✓	
39	B07NQN58ZC	✓		✓	
40	B07NQN58ZC	✓		✓	

41	B07NQN58ZC	✓		✓	
42	B07NQN58ZC	✓		✓	
43	B07NQN58ZC	✓		✓	
44	B07NQN58ZC	✓		✓	
45	B07NQN58ZC	✓		✓	
46	B005C2NA4E	✓		✓	
47	B005C2NA4E	✓		✓	
48	B005C2NA4E	✓		✓	
49	B005C2NA4E	✓		✓	

A.1. Calculation of Kappa Score

Number of Disagreement = 7/50

Table A.1. Kappa Score Calculation Table

Reviewer2 ➡	Correct	Incorrect
Reviewer1 ⬇		
Correct	43	6
Incorrect	1	0

P_0 => Number of Agreement/Total = 43/50 => 0.86, $P_{Correct}$ = 49/50 * 44/50 = 0.98* 0.88= 0.8624

$P_{Incorrect}$ = 1/50*6/50=0.02* 0.12= 0.0024, Pe = $P_{Correct}$ + $P_{Incorrect}$ = 0.8648

K= $(P_O - P_{e)}/ (1- P_e)$ => K = 0.86 – 0.8648/ (1- 0.8648) ==>K= -0.0048/0.1352 = -0.03 ~ 0

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